

Palate — an agentic marketing studio for a three-venue restaurant brand

The brief asked for a paper design of two agents. This submission answers **all four steps and both bonus questions** — and, because the clearest way to show how I think about agents is to let one run, every answer is backed by a **working product** you can sign into right now.

LIVE DEMO	https://palate-psi.vercel.app
SIGN IN	<code>owner@marlowandsage.com</code> · <code>palate2026</code>
GITHUB	https://github.com/ArchiusVuong-sudo/palate
STACK	Next.js 16 · Claude Agent SDK (Opus 4.8) · Gemini 3.1 image · Veo 3.1 video · Supabase
FULL AGENT RUNS	best experienced locally (<code>pnpm install && pnpm dev</code>) — long runs and the Chrome connection wizard exceed serverless limits; browsing, data and approvals work on the hosted demo.

Before designing anything — the questions I would ask the client

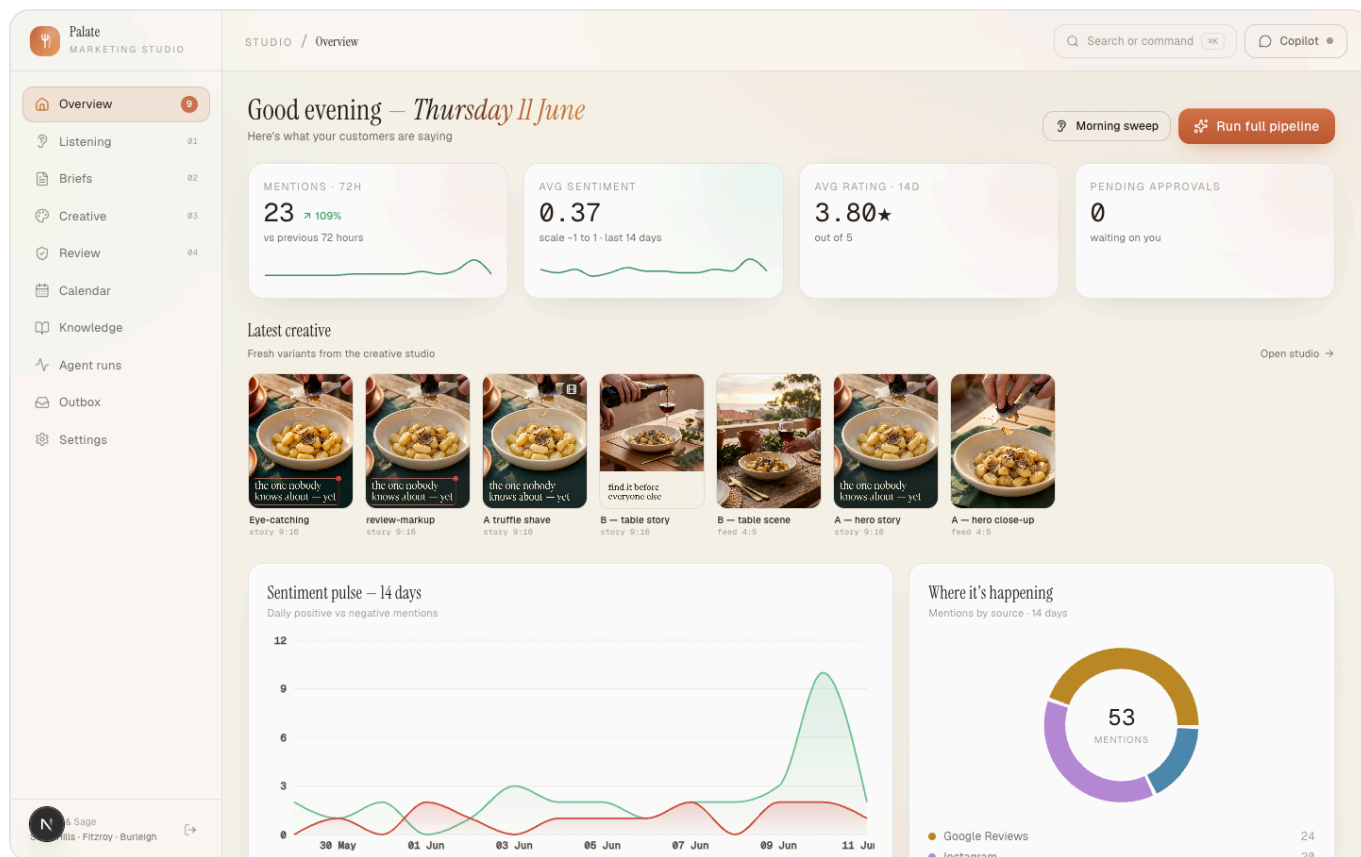
Two overwhelmed marketers don't need "an AI strategy"; they need hours back without losing control of their brand. Before proposing agents, I would sit with them for an hour and ask:

- **Where does the hour actually go?** Reading reviews, or deciding what to do about them? (The fix differs.)
- **What does "off-brand" mean here?** Show me three posts you killed and why — that becomes the review rubric.
- **What can the agent never do alone?** My assumption: anything customer-visible ships only after a human click.
- **Who owns the accounts?** Do we have admin on Google Business Profile and Meta, or do we need access sorted first?
- **What does success look like in 30 days?** Mine: the morning check drops from 60 minutes to a 2-minute digest, and the designer back-and-forth disappears for routine posts.
- **What's sacred?** For restaurants: food-safety complaints are never marketing material; staff names need consent; dietary claims must be literally true.

The product in this submission encodes those answers as defaults: human approval gates on everything outward-facing, a "knowledge filesystem" the team can read and edit in plain English, and hard guardrails for safety, consent and claims.

The shape of the system: one agent, four missions, many small tools

Rather than four separate black boxes, Palate is **one general agent** (Claude Opus 4.8 via the Claude Agent SDK) given four **mission prompts** — listening, briefing, creative, review — and a belt of **23 small, single-purpose tools** (query the database, save an insight, generate an image, request a human decision...). The agent composes the tools; the missions define the job; a shared, versioned **brand-knowledge filesystem** gives every mission the same memory. This matters commercially: when the client's workflow changes, you edit a mission in English — you don't re-engineer a pipeline.



The studio overview: live metrics, the latest creative, sentiment pulse, the agent's proposals — and a live feed of what the agent is doing right now.

Step 1 – Social Listening & Data Collection

Built as the “Listening” page + the *morning sweep* mission. Screenshot below.

1.1 What data sources would this agent monitor?

- **Google Reviews** for each of the three locations — the highest-stakes source (public ratings, food-safety signals).
- **Facebook**: comments on the brand page’s posts, visitor posts, and page mentions.
- **Instagram**: comments, @-mentions, photo tags, and story mentions for the brand account.
- **Worth adding once live** (not in the demo): TripAdvisor/OpenTable reviews, and a simple keyword watch (brand + dish names) on TikTok — where Australian food virality actually starts.

1.2 What tools or APIs collect from each source?

SOURCE	API / TOOL	NOTES FOR A NON-TECHNICAL TEAM
Google Reviews	Google Business Profile API	Official, free; needs the team’s Google account to grant access once. Pulls new reviews + ratings per location.
Facebook	Meta Graph API (page token)	One Meta developer app covers both networks. Webhooks push new comments in near-real-time.
Instagram	Instagram Graph API (via the same Meta app)	Requires a business IG account linked to the FB page — most restaurants already have this.
Analysis	Claude (agent) + structured tools	The LLM does sentiment, topic and dish extraction — no separate “sentiment service” to buy.

In the demo the three connectors run on realistic seeded data, and Palate ships a **connection wizard**: the agent co-drives a real Chrome window to the Google/Meta consoles, the human does the sign-ins, and the agent finds and stores the API keys — because for a 2-person team, *getting connected* is the real blocker, not the API choice.

1.3 What does the agent’s output look like?

Three layers, all produced by one morning run (about 4 minutes, autonomous):

1. **A 30-second digest** (chat + optional email) in a fixed order:  anything critical first →  trends with numbers →  opportunities →  bright spots → a “numbers” line (mentions/72h, avg sentiment, % negative, avg rating).
2. **Structured insights** saved to the database — each with a title, severity, the metric that triggered it, and *verbatim customer quotes as evidence*, so the team never has to trust a vague claim.
3. **A live dashboard** the agent itself publishes: sentiment-by-day chart, top topics, trending dishes with momentum, a 12-week activity heatmap, and a venue-by-venue radar.

Example from a real run on the seeded data: “ **CRITICAL** (*private, never post*): *Fitzroy served undercooked roast chicken again (9 Jun, 1★ — ‘pink and cold in the middle’). Same failure the kitchen*

was retrained on — this is a recurrence. Escalated for same-day temp-check verification; keep entirely off social.”

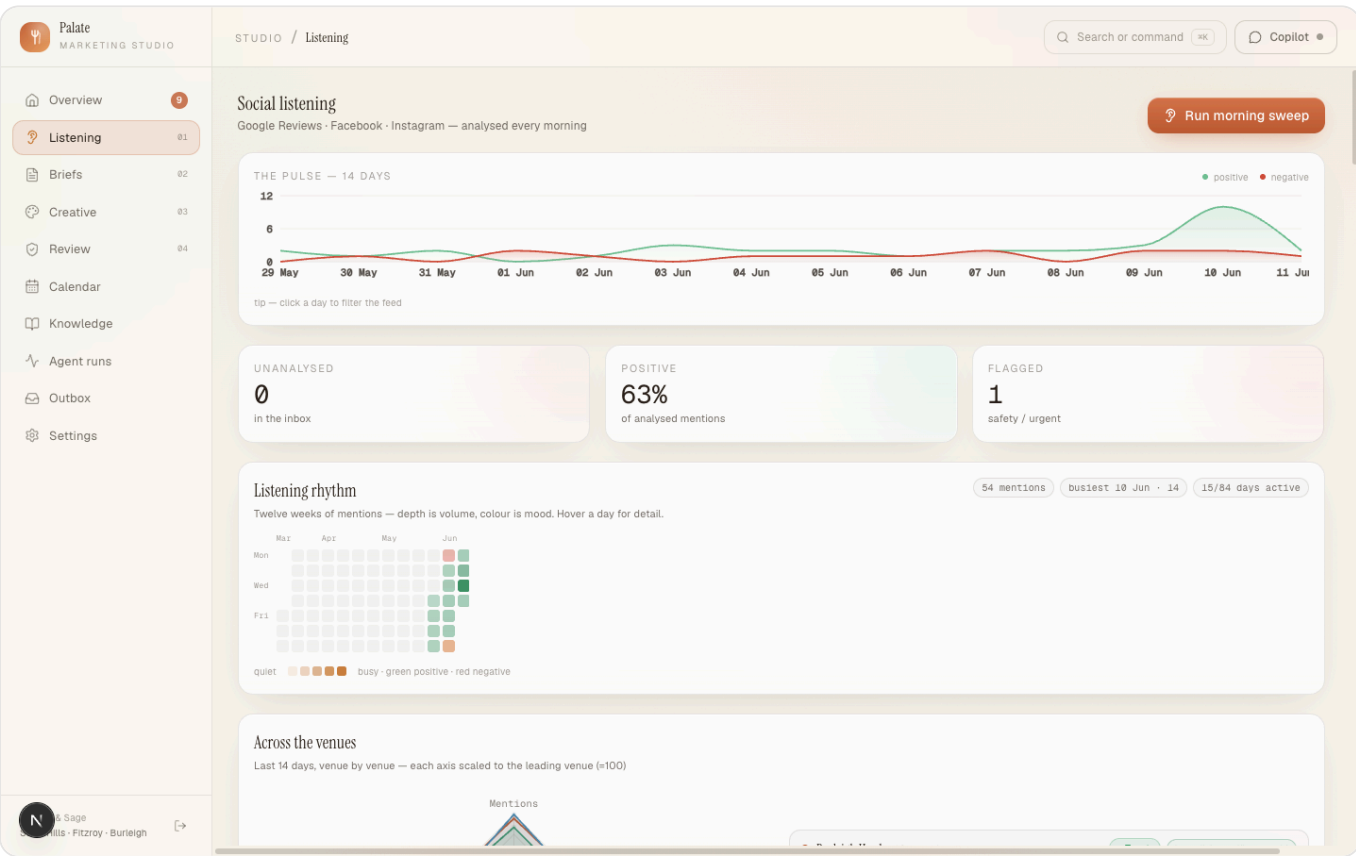
1.4 How does it decide “important” vs. noise?

Layered logic, in the mission prompt rather than brittle keyword rules:

- **Hard triggers** — anything health/safety, allergy or staff-conduct related is always critical and always private.
- **Change, not volume** — the agent compares against the prior 72 hours/14 days via SQL: a dish going 0→3 mentions matters; the 40th compliment about a dish we already know is loved doesn’t.
- **Actionability** — every insight must name an action; “sentiment is mixed” is banned by instruction.
- **Evidence or it didn’t happen** — each insight cites the underlying review IDs and quotes; fabrication is structurally impossible because insights without evidence quotes are rejected by the tool schema.
- **Caps** — 3–5 insights per day maximum, forcing prioritisation rather than a laundry list.

1.5 The one-sentence pitch to the marketing staff

“Every morning before you’ve finished your first coffee, it has read every review and comment across all three venues and hands you a 30-second digest — leading with anything that could hurt you, backed by the customers’ own words.”



Listening: 14-day pulse, 12-week rhythm heatmap, venue radar, the analysed feed, and the agent’s insights with evidence quotes.

Step 2 – Daily Content Briefing

2.1 What inputs does this agent take?

- The last 3 days of **insights** from Step 1 (with their evidence and metrics) and any still-open proposed actions.
- The **brand knowledge filesystem**: voice, visual style, hard guidelines, audience personas, location facts (which venue serves what, posting windows that work).
- **Recent briefs** (last 5) — so it never repeats yesterday’s angle or hits the same venue twice in a row.
- **Performance learnings** — a learnings file the agent itself appends to (e.g. “reels outperform statics ~3× for reach”).
- Optional **human focus** — the team can type “focus on the new winter menu” and the brief obeys.

2.2 What does a good brief contain for this team?

Specific enough that a designer asks *zero* follow-up questions:

FIELD	WHY IT KILLS THE BACK-AND-FORTH
Title + date + venue	One brief per day, one clear subject.
Objective	A business outcome (“convert truffle-gnocchi buzz into bookings”), not “make a post”.
Audience persona	Named persona from the knowledge files (“Date-night Dana”), with what she responds to.
Key message	Written in customer language, quotable directly in the caption.
Tone cues	Three adjectives from the voice contract (“warm, confident, quietly witty”).
Channels & formats	IG story 9:16 / feed 4:5 / FB — chosen, not left open.
Visual direction	References the visual contract: light, angle, palette, what’s in frame.
Copy notes / guardrails	Mandatory wording and forbidden claims for <i>this</i> post (e.g. “gnocchi contains gluten — never imply otherwise”).
CTA + schedule hint	What the reader should do, and the posting window that historically works.
Source insight IDs	Traceability: every brief can show <i>why</i> it exists.

2.3 Example system-prompt strategy (abridged from the running product)

You are **Palate**, the marketing agent for {brand}. Mission: create today's content brief.

1. Read the latest insights (last 3 days, with evidence) and open proposed actions.
Read learnings/content-performance.md and the last 5 briefs so you never repeat an angle.
2. Decide today's single best angle. Note in one line why it beats the runner-up.
3. Save ONE brief – specific enough that a designer would ask zero questions:
concrete objective, audience persona, key message in customer language, tone cues from brand/voice.md, channels, CTA, a visual_direction that references the visual contract, copy_notes with any mandatory wording, schedule_hint from audience windows.
4. Business rules: prefer momentum (rising mentions) over saturation (already selling out); never build content on safety complaints; honest-scarcity angles need manager

confirmation before publishing; rotate venues and angles across the week.
5. Close with the runner-up angle and when it would be the right play.

Note the structural tricks: *one* brief (forces prioritisation), “zero questions” as the quality bar, the runner-up requirement (makes the agent show its reasoning), and business rules in plain English the client can edit.

2.4 What business logic does it apply?

- **Momentum beats saturation** — a dish going 0→3 mentions with influencer pickup outranks the pavlova that already sells out nightly; posting the latter mostly amplifies what’s already full.
- **Respond vs. promote** — a trending *negative* topic (slow Friday service) is handled as review replies and an ops escalation, not as content; you don’t post your way out of an operations problem.
- **Friction check** — an angle needing manager confirmation (a scarcity claim) loses to an equally strong angle that doesn’t; the brief notes the blocked angle as “tomorrow’s play once confirmed”.
- **Safety wall** — critical insights are structurally excluded from briefing inputs.
- **Rotation** — venue and angle variety enforced across the trailing week.

In a real run, the agent picked the truffle-gnocchi launch over the pavlova for exactly these reasons — and wrote the pavlova up as the runner-up, ready for the day the manager confirms the batch-size claim.

2.5 “The brief doesn’t feel right for our brand” — what do I adjust?

I treat that sentence as **training data, not a complaint**. The fix is a loop, not a lecture:

1. **Localise the miss** — which field feels off: tone? angle? venue? visuals? (Each maps to a different knowledge file.)
2. **Capture the rule in their words** — the client tells the copilot in chat: “never call food ‘pillowy’, and feed captions must stay under 60 words.” The agent writes both into `brand/voice.md` as a new version (diffable, revertible) and confirms what it learned.
3. **Prove the loop closed** — this actually happened in the build: after learning those two rules, the agent later *caught an already-approved caption violating them* and flagged it for rewrite in its daily digest, unprompted.
4. **If it keeps missing** — the brief gains a “feedback” field and briefs regenerate against it; persistent misses mean the voice contract is underspecified, so we run a 30-minute voice workshop and rewrite it together.

Step 3 — Visual Content Generation

3.1 What tools, and why for F&B?

- **Gemini image generation (gemini-3.1-flash-image, “pro” escalation available)** — chosen for three F&B-specific reasons: (a) photoreal food texture without the “plastic AI sheen” that kills appetite appeal, (b) **reference-image conditioning**: we feed real photos of the actual dishes from a “brand library”, so the generated plating matches what a guest is actually served — for a restaurant, fidelity is a trust issue, not an aesthetic one, and (c) reliable 9:16 / 4:5 / 1:1 output at roughly **\$0.13 per image**.
- **Veo 3.1 (fast) for short video** — 6-second 9:16 clips with ambient audio (steam rising, truffle shaving), seeded from the chosen hero image so the story and feed stay visually consistent. Reels-style motion is where organic reach is.
- **Claude writes captions before any pixels** — copy is locked first because overlay text renders better when final, and because the caption *is* the brand voice; the image serves it.

3.2 Brief → image prompt (real example from the build)

Brief (abridged): *Winter truffle gnocchi — “the Burleigh dish nobody knows about yet”. Channels: IG story + feed. Visual direction: hero the dish per our visual contract; 45° hero shot, steam, glossy sauce, timber/terrace backdrop, golden-hour warmth, terracotta + cream tones.*

Generated image prompt (actual):

45° hero shot of hand-rolled potato gnocchi in a cream ceramic bowl, glossy brown-butter sauce, fresh black winter truffle being shaved over the top mid-shave (hand and shaver only, no posed face), visible steam, shallow depth of field with the bowl tack-sharp; weathered timber terrace table softly blurred behind; natural golden-hour warmth; terracotta and cream accents, eucalyptus-green linen; appetising, editorial food photography, no text.

The agent also attaches the brand library's **real reference photo** of the dish so Gemini matches the actual plating and crockery. For story formats it bakes in the overlay line in quotes (“the one nobody knows about — yet”) and the wordmark.

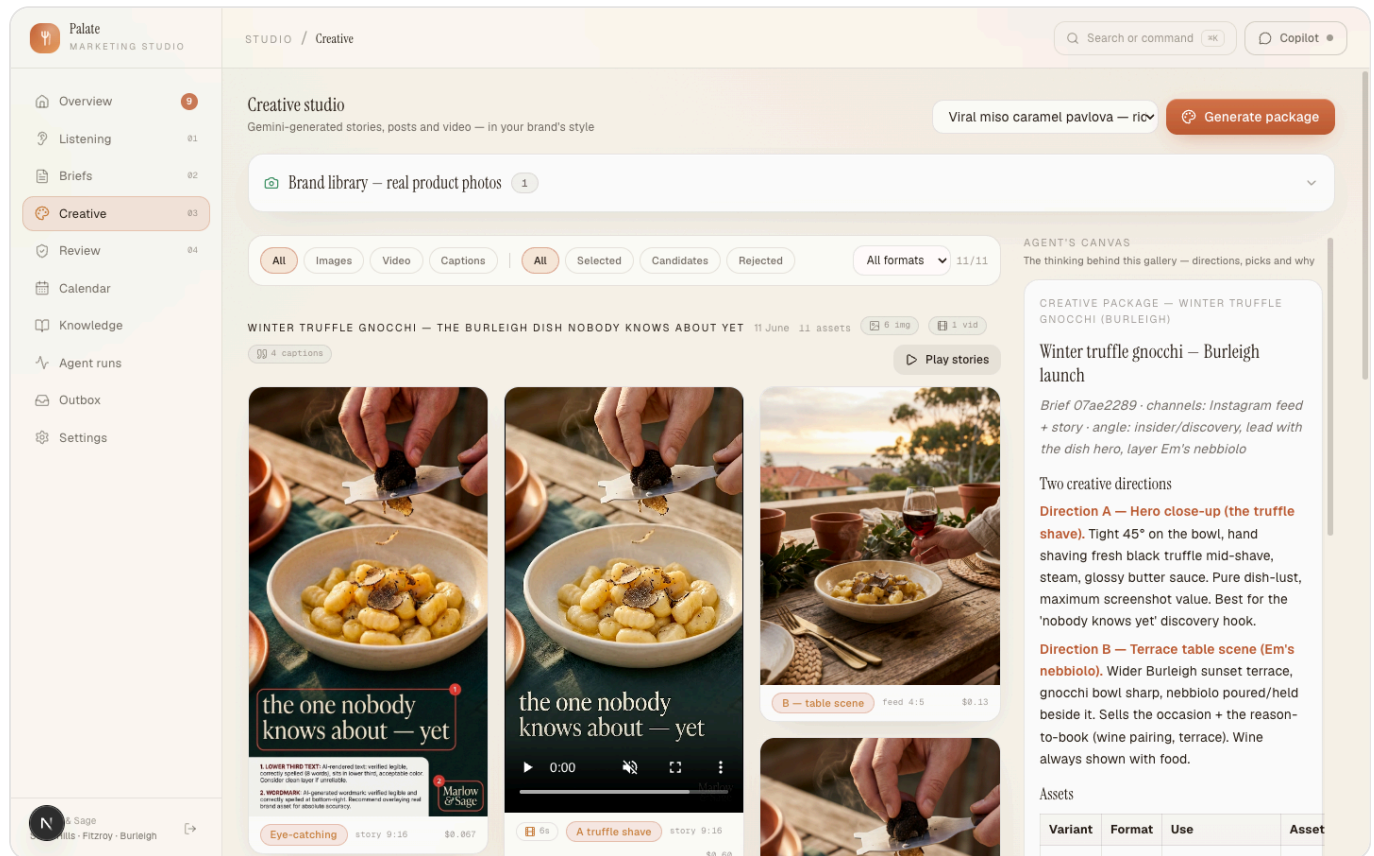
3.3 Style rules baked into every prompt

- Natural light only, warm golden tone; never studio-flash or neon. 45° or top-down angles.
- Brand palette accents in-frame (terracotta, cream, eucalyptus green); real textures (timber, linen, ceramic).
- Steam/moisture cues for hot dishes; shallow depth of field; the dish is always the hero.
- People: hands and service moments only; no posed faces (consent + timelessness).
- Text: either “no text” or the exact overlay copy in quotes — never improvised typography.
- Alcohol only ever shown alongside food (responsible-service rule in the guidelines file).

These live in `brand/visual-style.md` — the team edits a text file, not a prompt template, and every future image obeys.

3.4 What does the marketing staff actually receive?

A **package per brief**, streamed live into a gallery as it's made (≈5 minutes, ~\$1 all-in): 2 creative directions × each required format (4–6 images), 2 caption variants per channel with hashtags, and one 6-second video — plus the agent's own recommendation of the hero combination and why. The gallery has one-click select/reject, an "Instagram preview" that renders any candidate inside a phone mock-up, a story player, and a refine box ("make the sauce glossier") that edits the image in place.



The creative studio: brand library of real dish photos, variant gallery, captions, and the agent's canvas notes to the right.

3.5 What still needs a human, and why

- **Choosing the hero** — taste and context (what's happening in the venue this week) stay human; the agent recommends, never decides.
- **Visual truth-check** — does the render match what guests are served? AI food can drift; only staff know today's plating.
- **Rendered-text inspection** — AI overlay text can garble; notably, our review agent learned this itself and started using its annotation tool to visually verify overlay text rather than trusting the prompt.
- **Real-photo moments** — generation fills the content calendar; it should never replace genuine staff/guest photos, which are the trust currency of a local restaurant.

Step 4 — Brand Consistency Review

4.1 What must a “brand guideline” contain — and what do I ask the client to prepare?

Four short documents, written with them in a workshop (I draft, they correct — far easier than asking marketers to write specs):

- **Voice contract** — 3 tone adjectives, sentence-case rule, AU spelling, emoji policy, a banned-word list (“foodie”, “nom”, ... later “pillowy”), exclamation limits, caption length caps.
- **Visual contract** — the Step-3 style rules above, plus logo/wordmark usage.
- **Hard guidelines** — the non-negotiables: staff can be named only from a consent list; no dietary claims unless literally true; no unverified scarcity claims; alcohol only with food; price mentions policy; food-safety topics never on social.
- **Ground truth** — personas, per-venue facts (menus, hours, what’s exclusive where), so claims can be checked.

4.2 How are guidelines represented for the agent?

Plain-English files in a versioned knowledge filesystem, injected into the prompt — deliberately not a vector database. The corpus is ~7 small documents; it fits in context whole, so retrieval can’t silently drop the one rule that mattered. The wins of this representation: the *client can read and edit their own “model”*; every change is a new version with a diff and one-click restore; and the agent itself appends learned rules (tagged as agent-authored) which the team can audit. Hard rules additionally get **server-side enforcement**: publishing tools literally refuse to run without an approved human decision attached — prompts persuade, but code guarantees.

4.3 What does the agent check? (the rubric)

1. **Voice & tone** — banned words, emoji/exclamation rules, AU spelling, sentence case, length caps.
2. **Visual style** — light/angle/palette per the visual contract, judged by *looking at the image*, not the prompt.
3. **Guideline compliance** — the hard rules: consent list, dietary claims, scarcity honesty, alcohol-with-food, pricing.
4. **Message accuracy** — every factual claim cross-checked against ground truth (right venue, dish actually on the menu there).
5. **Audience fit** — right persona and platform convention (a story hook is not a feed caption).
6. **Legibility of rendered text** — overlay copy and wordmark verified visually via the annotation tool.

Each scores 0–100 with a one-line note, so “feels off-brand” becomes a number the team can argue with.

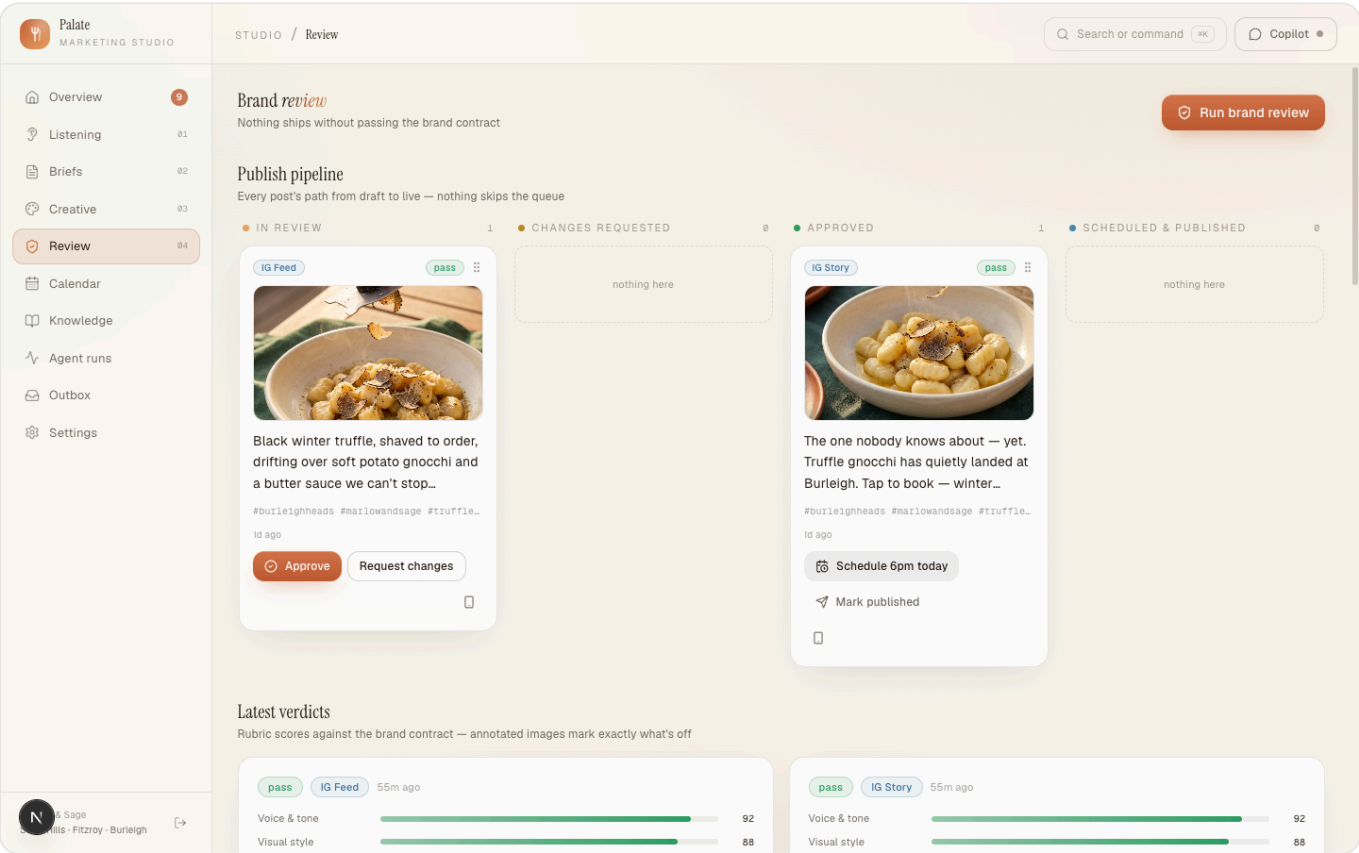
4.4 Possible outputs — pass, flag, reject

VERDICT	TRIGGER	WHAT HAPPENS
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Pass	All criteria ≥ 75 and zero hard-rule breaches	Scorecard saved with strengths noted $\rightarrow$ goes to the human approval gate (pass $\neq$ published; a person still clicks).
Flag	Any 50–74, or a fixable issue	Precise fixes listed; visual problems are <i>drawn on the image</i> (annotated boxes); one retouch attempted, then re-reviewed.
Reject	Any hard-rule breach, or any score < 50	Plain-English explanation; post moved to “changes requested”; the lesson appended to the learnings log so the same mistake fades.

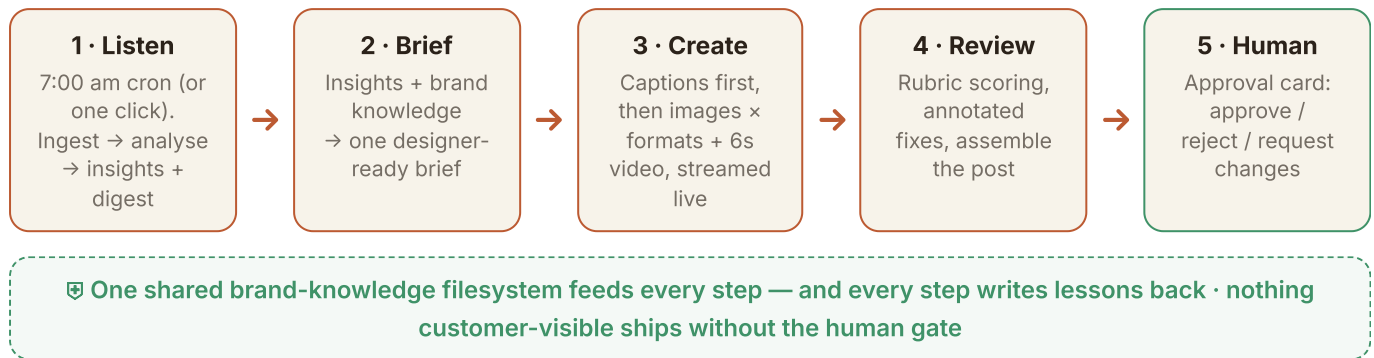
4.5 Preventing too-strict / too-lenient

- **Asymmetric instruction** — “strict on hard rules, lenient on taste.” Only rule breaches can reject; style nits can only flag. This single line removes most false blocks.
- **Numbers, not vibes** — explicit verdict thresholds make the agent’s strictness *tunable* (move the pass bar, don’t re-prompt).
- **The human gate is the safety valve** — an over-strict reviewer can’t block a human who disagrees; the approval card shows the scorecard and the human decides. Every override is feedback.
- **Overrides become calibration** — rejections the team overrides, and approvals the team reverses, are written into the learnings file; the reviewer drifts toward the team’s real taste instead of an imagined one.
- **Spot-audit** — periodically re-score a sample of human-approved published posts; sustained disagreement means the contracts need a rewrite, not the agent.



The review board: publish pipeline as a kanban (nothing skips the queue), rubric scorecards, and human approval as the final gate.

Bonus A — The end-to-end flow



Triggers. Three ways in, same machinery: the 7:00 am *autopilot* cron runs the whole cycle; any single step can be run on demand from its page ("Generate today's brief"); and the chat copilot can invoke any step conversationally ("draft a reply to the slow-service review"). Steps hand over through the database — each phase's output is a persisted, human-editable artifact (insights → brief → assets → post), not an in-memory message. That makes the pipeline *interruptible and resumable by design*: a human can edit the brief before creative starts, or reject one image without rerunning listening.

If a step fails (API outage, budget cap, timeout): the run is marked failed with its full tool-by-tool audit trail kept; everything already produced persists (a failed creative run that made 3 of 5 images keeps the 3); nothing downstream auto-publishes because the human gate never fires; the team sees it in the run log and the daily digest, and a "Continue" button resumes the agent's session with full context rather than starting over. Failures degrade to "the team does today the old way", never to "something wrong went out".

A real unscripted run of the whole loop: during the build, the morning cron fired on its own. It analysed overnight mentions, noticed the already-approved feed caption violated two rules the owner had taught the agent the day before (the word "pillowy"; a 60-word cap), wrote a compliant replacement, and **blocked at the approval gate** asking permission to swap it — citing rubric scores and the rules by name. That's every step, plus the learning loop, working together without a human kick-off.

Bonus B — "Can we have one button that does everything automatically, without us checking?"

What I'd actually say to the client:

"Almost — and the 'almost' is what protects you. Palate has that button: Autopilot runs the whole cycle every morning at 7. By the time you open your laptop, the listening is done, today's brief is written, the visuals are made and brand-checked. What it deliberately won't do is *publish* without you. The last click — about two minutes of your day — stays human, for three reasons.

"First, the stakes live at the edges. Last week the system caught an undercooked-chicken review. A fully automatic pipeline optimising for engagement might have amplified food content that day; a human

glance knows the difference instantly. You're not checking its grammar — you're supplying the judgement only you have. Second, accountability: when something goes out under Marlow & Sage's name, a person chose it — 'the AI posted it' is not an answer you ever want to give a journalist or a regulator. Third, your two minutes are the training signal: every approve, reject and edit teaches the agent your taste, so the checking gets faster every week — it already caught one of its own captions breaking a rule you'd taught it the day before. And honestly, this isn't me being conservative: every serious AI marketing platform — Jasper, Klaviyo, HubSpot — ships exactly this human-approval model. If the review burden ever stops trending toward zero, that's a bug in my system, not a reason to remove the gate."

Beyond the brief — what the working build demonstrates

CAPABILITY	WHAT IT SHOWS ABOUT AGENT DESIGN
Human-in-the-loop that actually blocks	The agent's <code>request_approval</code> tool suspends the run mid-thought until a human clicks; gated actions (publish, email) are enforced server-side — the model can't talk its way past them.
A memory the client owns	Versioned knowledge files with diffs, history and restore; the agent appends its own lessons (tagged), the team audits them. The "pillowy" rule went chat → voice.md v2 → enforced next day, autonomously.
Conversational copilot with continuity	Persistent chat resumes the same agent session; any finished run can be continued ("ask a follow-up") with full context — verified across a 19-hour gap.
Connection wizard (agent + browser)	The agent co-drives a visible Chrome window through Google/Meta consoles to fetch API keys; humans do every sign-in via approval gates. Onboarding as an agentic feature (run locally).
Operations cockpit	Content calendar with drag-to-reschedule, live agent activity feed, run audit trails with per-run cost, autopilot toggle, command palette, email digest outbox.
Observed economics	Measured on Opus 4.8: morning sweep ≈ \$0.76 · brief ≈ \$0.40 · creative package ≈ \$0.50 + ~\$0.13/image + ~\$0.60/video · brand review ≈ \$0.98 · copilot turns \$0.06–0.29. A full daily cycle ≈ \$2–3/day (~A\$100/month) against ~1.5–2 staff-hours/day saved.

Honest limitations & the 30-day path to production

- **Connectors are seeded** in the demo; the Graph/Business Profile wiring slots into the existing connection cards (the wizard already handles credential capture). Week 1–2.
- **The learning loop is event-driven, not metric-driven yet** — next step is ingesting post performance so briefs and reviews are informed by what actually worked (the calendar and review rubric are built to receive it). Week 2–4.
- **Hosted demo limits** — serverless caps long agent runs and the in-memory approval bridge prefers a single instance; the production answer is a small worker for runs (the code is already structured for it). Local runs have no such limits.
- **Costs are capped** per run (\$8 hard budget) and audited per tool call — the client sees every dollar.

REVIEWER

QUICKSTART Sign in at palate-psi.vercel.app (`owner@marlowandsage.com` / `palate2026`) → Overview shows the live studio.

5-MINUTE TOUR Listening (heatmap, insights with quotes) → Briefs (chips + “why this angle”) → Creative (gallery, phone preview, agent’s canvas) → Review (kanban + scorecards) → Knowledge (open `brand/voice.md` → History → see the agent’s v2 diff) → Runs (audit trails, costs, “Continue”).

FULL AGENTIC

EXPERIENCE `git clone https://github.com/ArchiusVuong-sudo/palate && npm i && npm dev` with the provided `.env.local` → click “Run full pipeline” and watch the dock; answer the approval card it raises.

Palate — built end-to-end (product, agents, deployment, this report) as the AT Solutions take-home by Khanh Vuong, June 2026. Demo data is fictional; models: Claude Opus 4.8 (agent), Gemini 3.1 Flash Image, Veo 3.1, via the Claude Agent SDK.